MOBILE DJ / PRODUCT BLUEPRINT

DOCUMENT 02 OF 19

Professional product requirements

A new touch-first performance instrument.
Professional audio, liquid glass and mechanical craft.



DESIGN HANDOFF BASELINE - PLANNING ONLY

29 October 2026 | Crystal Current selected; S26 Ultra test target

Independent of the previous Mixline preview. Targets are not measured results.

Version: 1.0 **Date:** 29 October 2026 **Status:** Design handoff baseline - implementation and device validation pending **Product:** New mobile DJ application; no Mixline reuse **Platform direction:** Android first; Galaxy S26 Ultra is the confirmed first test device (current software reported, not inspected) **Owners:** Product lead, audio lead, mobile lead, design lead and QA lead

1. Requirement contract

The intended product is a complete mobile performance instrument: quick to start, deep enough for professional preparation and live creativity, and dependable during a set. The user confirmed that external-controller integration is excluded. External MIDI/HID DJ controllers and their mapping therefore do not belong in this baseline. DVS/timecode control is proposed as part of that exclusion and should be confirmed. USB audio interfaces and dedicated DJ splitters are explicitly allowed. Microphone functionality is selected; exact supported microphones, speaker routes and external displays still require a hardware matrix.

Every “shall” below expresses a proposed technical acceptance contract for the selected feature scope. This handoff is ready for design; implementation and test evidence are subsequent work. Two decks are the proposed default visible workspace; four-deck performance remains a selected advanced capability. Video mixing, karaoke, broadcasting and third-party plugin hosting are selected requirements. Lighting is excluded. Specific mobile formats, compatible plugins and device tiers remain to be proven, and a missing dependency must be reported rather than silently deleting a selected feature.

The owner confirmed a local Android version first and AWS expansion later. No required LAN/cloud server, AWS account or cloud trial belongs in the local-version dependency chain. The future AWS target is about US$35/month excluding ParDay AI-provider charges; it is a planning target, not spending authorization or a hard cap. Online source operations, broadcasting and optional AI remain independently network-dependent. Preserve the requested local link-to-file feature and report source-specific feasibility gaps rather than silently replacing it with a mandatory cloud service.

Professional capability is measured through completed workflows, accurate timing, audible results, usable controls and recovery. A control that animates without driving the engine does not satisfy a requirement. An engine function without a usable and accessible control does not satisfy the corresponding product workflow.

2. Gate vocabulary

G0 establishes the decision baseline. G1 proves audio feasibility. G2 is a working vertical slice. G3 is core performance beta. G4 completes advanced performance. G5 covers source/provider and ParDay integrations. Cloud-specific G5 cases apply only when the later optional AWS extension is evaluated. G6 establishes release candidacy. G7 is the personal installed release. G8 is maintenance and optional community readiness. G3 is not the full requested final product. The user has selected all discussed professional modules except lighting for the complete target. G7 cannot be described as that complete product with selected modules missing, but it does not wait for an AWS extension. Exact provider coverage, compatible plugin formats and certified device combinations remain unresolved implementation constraints.

In the matrix, a gate is the earliest evidence checkpoint. All selected release requirements must also pass regression at G6. QA identifiers are stable test-case references for the quality plan; tests must assert observable behavior rather than repeat implementation code.

3. Functional requirement matrix

ID	Proposed requirement	Acceptance evidence	Owner / first gate
FR-001	Core local playback shall work offline without an app account.	QA-FR-001: fresh guest session imports and mixes two authorized local tracks in airplane mode.	Mobile + Audio / G2
FR-002	Import shall use user-selected files/folders and retain durable access or make a verified local copy.	QA-FR-002: restart, moved file and revoked access produce correct available/missing states without lost metadata.	Data / G2
FR-003	Supported formats shall be enumerated by codec/container/bit depth/sample rate, initially evaluating MP3, AAC/M4A, WAV, AIFF, FLAC, Ogg/Vorbis and Opus.	QA-FR-003: approved corpus decodes correctly; malformed/unsupported files fail safely. Final format set follows G1 decoder/license review.	Audio + Security / G3
FR-004	Search shall filter title, artist, album, BPM range, key, tags, availability and played state.	QA-FR-004: exact fixtures return expected results; search remains usable during playback on the declared library-size tier.	Data + Design / G3
FR-005	Crates/playlists, folders, smart rules, ratings, colors, comments, favorites and history shall support set preparation.	QA-FR-005: reorder, rename, duplicate detection and saved filter behavior survive restart.	Data / G3
FR-006	BPM, downbeat, key, waveform and loudness analysis shall run outside the real-time callback and cache versioned results.	QA-FR-006: analysis progress/error is visible; playback does not depend on completed optional analysis.	DSP + Data / G3
FR-007	Variable-tempo beatgrids and phrase/energy hints shall distinguish detected results from user corrections.	QA-FR-007: live-drumming/tempo-change fixtures retain meaningful bar positions; low-confidence results are not presented as certain.	DSP / G4
FR-008	Users shall correct downbeats, beat positions, local tempo, half/double BPM and grid alignment, then lock edits.	QA-FR-008: loops and sync follow corrected maps; reanalysis cannot overwrite locked edits silently.	DSP + Design / G3-G4
FR-009	Overview and detailed waveforms shall be derived from media, with cue/loop/grid overlays and explicit zoom.	QA-FR-009: transient positions match reference audio; overview seeking and display clocks stay consistent.	Audio + UI / G3
FR-010	Each deck shall provide load, play/pause, stop, temporary cue, nudge, position and elapsed/remaining time.	QA-FR-010: independent state and multi-touch transport actions produce documented audible results.	Audio + UI / G2
FR-011	Each track shall support at least eight named/color-coded hot cues plus saved memory markers.	QA-FR-011: cues persist, trigger at intended musical positions and distinguish set/delete/trigger actions.	Audio + Data / G3
FR-012	The engine shall support four deck identities; four simultaneous decks shall be enabled only for a certified workload tier.	QA-FR-012: switching focus never changes transport or assignment; four-deck workload passes performance gates.	Audio + UI / G4

ID	Proposed requirement	Acceptance evidence	Owner / first gate
FR-013	Touch jog/scratch shall support vinyl mode, nudge, directional movement and defined release behavior.	QA-FR-013: scratch/release tests measure audio response; pointer cancellation cannot leave a deck held.	Audio + UI / G3
FR-014	Waveform seek/scrub shall be a first-class action distinct from jog scratching.	QA-FR-014: precise seek, coarse navigation and accidental-gesture protection work with track running or paused.	Audio + Design / G3
FR-015	Tempo, pitch bend, key lock and independent key shift shall expose ranges and reset controls.	QA-FR-015: reference playback verifies pitch/time behavior and smooth parameter changes across the approved range.	DSP / G3
FR-016	Manual matching and optional sync shall coexist with visible leader/follower and phase state.	QA-FR-016: sync off returns predictable manual control; invalid grids cannot silently force a bad alignment.	Audio + DSP / G3
FR-017	Quantization shall support selectable musical resolution with explicit enabled state.	QA-FR-017: cue, loop and pad commands meet the documented event scheduling rule; unquantized commands are not delayed to a beat.	Audio / G3
FR-018	Manual/auto loops, loop in/out, reloop, loop roll, beat jump and saved loops shall be supported.	QA-FR-018: boundaries, negative jumps, track edges and changing-tempo maps are exercised without invalid reads.	Audio + DSP / G3-G4
FR-019	Slip and reverse shall preserve or replace timeline position exactly as the chosen mode specifies.	QA-FR-019: scratch/loop/cue during slip returns to the calculated musical position; release is click-controlled.	Audio / G4
FR-020	Mixer shall include trim, channel faders, EQ/isolator, filter, assignable crossfader sides/bypass and curve/cut-in.	QA-FR-020: signal-path tests verify gains, bypass, center behavior, resets and audible smoothness.	DSP + UI / G3
FR-021	Channel/master metering, peak hold, clipping indication and master protection shall reflect the actual signal.	QA-FR-021: calibrated fixtures exercise overload; meter location and limiter latency are documented.	DSP / G3
FR-022	Independent cue/PFL, cue/master blend and headphone gain shall work only on valid output routes.	QA-FR-022: isolated cue never leaks to main; split-mono mode clearly trades stereo for independence.	Audio + Platform / G3
FR-023	Route setup shall identify main/cue channels, sample rate and unsupported combinations.	QA-FR-023: connect/disconnect/reconfigure on supported hardware yields truthful routes and safe fallback.	Platform / G3
FR-024	FX shall offer essential echo, delay, reverb and filter plus advanced modulation/distortion, assignment, wet/dry and tempo/free rate.	QA-FR-024: impulse/reference tests verify routing, bypass and tails; no timing depends on UI frame rate.	DSP / G3-G4
FR-025	FX XY control and macros shall show parameter mapping, hold/latch mode and reset behavior.	QA-FR-025: two-axis gestures, multi-touch and cancellation map predictably; reset cannot leave hidden wet gain.	UI + DSP / G4

ID	Proposed requirement	Acceptance evidence	Owner / first gate
FR-026	Performance pads shall have explicit cue, loop, roll, jump, slicer, sampler, pitch-play and stem modes where supported.	QA-FR-026: changing bank/mode cannot trigger a hidden action; active mode and deck remain legible.	UI + Audio / G4
FR-027	Sampler/sequencer shall support at least 16 slots, banks, trigger modes, editable step patterns, per-step velocity/mute, length/resolution, swing and quantized start/stop/change.	QA-FR-027: approved voices and saved banks/patterns survive restart; repeated patterns, tempo changes, bar-boundary edits, mute/stop and takeover follow engine time.	Audio + Data / G4
FR-028	Prepared stems shall support aligned vocals/drums/bass/other with mute/solo/gain and readiness status.	QA-FR-028: sum/alignment reference, cache integrity and unavailable-stem fallback are verified on authorized media.	DSP + Data / G4
FR-029	Live separation shall be conditional on source permission, model license and measured device capacity.	QA-FR-029: refused activation is understandable; overload does not disrupt already playing full-mix audio.	DSP + Platform / G4
FR-030	Microphone/aux input shall expose permission, route, gain, mute, monitoring and talkover/ducking.	QA-FR-030: feedback-aware defaults and supported input/output latency are verified on actual hardware.	Audio + Platform / G4
FR-031	Mix recording shall provide format choice, timer, storage estimate, markers, safe finalization and partial recovery.	QA-FR-031: low storage/process interruption recover valid segments; playback remains independent of writer failure.	Audio + Data / G3
FR-032	Queue/automix shall show next readiness, transition settings, reorder, skip, cancel and immediate manual takeover.	QA-FR-032: unavailable next track is never treated as ready; takeover causes no surprise transport restart.	Audio + Product / G4
FR-033	Preview and track preparation shall remain available while the current set plays.	QA-FR-033: browsing/search/editing never obscures required transport, output and fader access.	Design + UI / G3
FR-034	Session state shall persist deck assignments, cues, relevant controls, queue and history at defined checkpoints.	QA-FR-034: recovery presents a truthful saved state and requires deliberate audible resume after a crash.	Data + Audio / G3
FR-035	Live protection shall prevent unintended audible-deck replacement and destructive reset.	QA-FR-035: load protection, live lock, quick fade and escape actions are tested under realistic touch errors.	Design + Audio / G3
FR-036	Source capability resolution shall gate analysis, caching, DSP, stems, recording, export and broadcast independently.	QA-FR-036: unknown/expired permission fails closed; provider restrictions are enforced below the UI.	Integrations + Security / G5
FR-037	Background playback, audio focus and interruption behavior shall be explicit per operating system.	QA-FR-037: calls, notifications, lock screen, route loss and system termination follow approved state transitions.	Platform / G3
FR-038	Day/night, high contrast, reduced transparency/motion and color-independent identities shall be supported.	QA-FR-038: critical controls and labels remain usable in each mode; screen-reader/precise-value alternatives exist.	Design + Accessibility / G3

ID	Proposed requirement	Acceptance evidence	Owner / first gate
FR-039	Portrait/landscape and compact/expanded layouts shall preserve engine state and pointer safety.	QA-FR-039: rotate during playback/gesture without audio reset, cue leakage or unintended fader movement.	UI + Platform / G3
FR-040	Accounts shall be optional for the offline core; cloud functions shall disclose the data they require.	QA-FR-040: expired app login cannot disable authorized local performance unexpectedly.	Product + Security / G3
FR-041	Backup/export shall include portable metadata, version information and explicit source references.	QA-FR-041: round-trip restoration restores user work; restricted media is not copied as a side effect.	Data / G4
FR-042	Diagnostics shall show meaningful audio-route, storage, battery/thermal and source readiness states.	QA-FR-042: simulated faults produce actionable messages; telemetry omits track contents/tokens by default.	Platform + Security / G3
FR-043	Bundled audio, fonts, icons, textures and models shall have documented rights and source versions.	QA-FR-043: every shipped asset resolves to an approved manifest entry and required attribution.	Assets + Release / G6
FR-044	Provider loss shall not invalidate the local engine, user-owned library or metadata unnecessarily.	QA-FR-044: provider unavailable/revoked while local decks play leaves those local workflows operational.	Integrations / G5

4. Selected specialist and assistant requirements

FR-045 - Video mixing: selected. Support authorized video decks, visual crossfades, linked/unlinked audio-video behavior and visual effects, with external display on named supported routes. Document 15 defines the codec, audio-clock, seek, visual fallback, thermal and output contracts. Owner: Audio/Video + Platform. QA-FR-045 and the AV cases in Document 15. First feasibility G1; integrated evidence G4; release regression G6.

FR-046 - Karaoke/broadcast: both selected, with separate submodule acceptance. Karaoke includes singer rotation, authorized timed lyrics/media, mic integration and supported output. Broadcasting includes eligible mix encoding, explicit destinations/start/stop, protected credentials, visible connection state and independent failure handling; camera/visual overlays require their own resource profile. Document 15 defines these contracts. Personal/private practice does not grant internet broadcast rights. Owner: Audio/Video + Integrations. QA-FR-046 plus AV cases; G4/G5, regression G6.

FR-047 - Third-party plugin hosting: selected; lighting is explicitly excluded from this requirement and the product. Support at least one genuinely third-party, Android-compatible effect format/path after feasibility, distribution and licence review. Built-in effects alone do not satisfy hosting, and desktop plugin binaries are not presumed compatible. Versioned ABI, latency reporting, resource budgets, failure isolation, preset recall and allowed installation path are required. Document 15 provides the feasibility and acceptance plan. Owner: Architecture + Audio + Security. QA-FR-047 plus AV cases; first feasibility G1, integration G4, release G6.

FR-048 - ParDay assistant: selected. Provide a provider-neutral assistant for eligible local-library search, crate preparation, readiness diagnosis, transition proposals and optional rehearsal and selected armed assisted performance. The local validator enforces source permissions, user authority, fresh deck/session state and bounded actions; a model never drives the audio callback. Option C is confirmed: preparation assistance plus explicitly armed bounded transitions/automix and immediate manual takeover. Before a concrete plan is armed, live behavior remains read/propose. Owner: AI + Product + Security + Audio.

QA-FR-048, AI-001 through AI-024 in Document 13; deterministic harness G2, cloud integration G5, release G6.

FR-049 - Music discovery and acquisition: selected in-app link-to-file acquisition, including the requested unofficial Spotify/YouTube downloader capability. Provide Find a playable version across catalogs, local media and acquisition adapters. Evaluate an own/self-hosted extractor and third-party integration as concrete candidates; Document 16 separates technical verification, provider access, content rights and distribution review. An official API is not a universal technical prerequisite. Preserve recording/version identity, actual media origin, provenance and operation-specific permissions. A URL/search result or streaming offline cache is not treated as an exportable music file. No named adapter is declared working or release-cleared before evidence exists. Owner: Integrations + Data + Security. QA-FR-049, SRC-01 through SRC-12 in Document 14, and DLA-01 through DLA-20 in Document 16; local import G2, admitted provider workflows G5, release G6.

5. Cross-cutting acceptance rules

Each test result records device/OS, audio route, sample rate/buffer, workload, media corpus, thermal condition and software versions. Performance targets live in the audio/quality documents; a single fast demonstration does not certify a platform. Differences between devices and source providers must be visible in the capability matrix.

Library editing, artwork decoding, analysis, stem generation, network work and recording writes must not block the real-time audio callback. UI animation follows the audio clock; it never becomes the clock. Operations entering the engine include stable deck identity and ordering so rapid gestures cannot affect the wrong deck after focus changes.

A full release review traces every selected FR to implementation, test evidence, known limitations and user documentation. A feature is complete only when success, loading, unavailable, error, cancellation and recovery states are implemented. Nothing is declared finished because an AI-generated screenshot shows the control.

6. Research basis

Official feature research informs category coverage; the requirements above are original proposals. Platform parity and partner permissions must be checked independently.

- Algoriddim, djay Pro capabilities.
- Serato, DJ feature comparison.
- AlphaTheta, rekordbox mobile and mobile FAQ.
- Native Instruments, Traktor Pro.
- VirtualDJ, professional feature categories.

Remaining implementation and later-phase decisions: capture the S26 Ultra's exact installed build/variant and audio route; qualify broader devices, source coverage and compatible formats/plugins; choose a ParDay provider and separate AI spending limit before paid model use; resolve optional voice and source-publication/licence details when needed. Team, development budget and schedule remain uncommitted. AWS account, region, service and current quote are deferred to expansion; the planning target of about US$35/month excluding AI is already confirmed. Establish the on-device acquisition baseline now and evaluate AWS later before choosing an expansion runtime. C Crystal Current, Galaxy S26 Ultra first, ParDay option C, personal-tool delivery and local-first sequencing are confirmed. No local host is selected or required. Future open-source sharing remains optional. Previously answered questions must not be asked again or reset to proposed status. No unanswered item blocks the first design work package in Document 19; feature implementation and device capacity remain unverified.